Vusumuzi Nkosi

Data Scientist · Forecasting · Fraud & Risk Analytics · Experimentation

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SUMMARY

Data scientist specialising in **forecasting, fraud and risk analytics, and experimentation**. I start from the decision a team has to make, prove each model beats a simple baseline, and explain the result in plain language, a skill sharpened over nine years of teaching mathematics. Recent work: **3.1% error** forecasting South Africa's electricity demand a week ahead (and 2.4% a day ahead, against Eskom's own published 2.3%), **98% of fraud caught** within a team's daily review capacity, and **82% of revenue at risk protected** by a ranked churn call list. Completing a PGDip in Data Science (STADIO, NQF 8); seeking data scientist or data analyst roles in South Africa or remote.

TECHNICAL SKILLS

Programming: Python, SQL (CTEs, window functions, DuckDB), R (basic)

ML & Modelling: scikit-learn, XGBoost, gradient boosting, random forest, logistic regression, K-means clustering, feature engineering, cost-based thresholds, time-based and walk-forward validation, imbalanced data

Statistics & Experimentation: hypothesis testing, A/B testing, McNemar's test, confidence intervals, power analysis, regression analysis, time-series forecasting, SciPy, statsmodels

Data & Visualisation: pandas, NumPy, Matplotlib, Seaborn, Streamlit, Tableau, Excel

Tools & Platforms: Git, GitHub, Jupyter Notebook, Google Colab, AWS SageMaker, Prompt Engineering, Generative AI

PROJECTS

Eskom Demand Forecasting — Python, scikit-learn, gradient boosting

github.com/vusumuzinkosi/eskom-demand-forecasting

- Forecast South Africa's hourly electricity demand a week ahead from 4.5 years of Eskom data (39,600 hours), using calendar, public-holiday and lagged-demand features with 52 weeks of walk-forward validation.
- Reached 3.1% MAPE a week ahead, 15% better than a “same as last week” baseline, and 2.4% a day ahead against Eskom's published 2.3%, without weather data.

Fraud Detection: A Daily Review Queue for Mobile Money — Python, scikit-learn

github.com/vusumuzinkosi/fraud-detection-mobile-money

- Built a daily review queue for a six-analyst team from 2.6 million mobile money transactions, with a time-based split and only information known at transaction time; found and removed a leaking balance column.
- Reviewing the top 300 transactions a day catches 98% of fraud cases (89% of the money) with 87% of flags genuine, where the existing rule caught 1 fraud in 4 days.

Churn Prediction: Revenue at Risk — Python, scikit-learn, Streamlit

github.com/vusumuzinkosi/churn-prediction-revenue-at-risk · Live demo: vusumuzinkosi-churn-planner.streamlit.app

- Trained logistic regression and random forest models with a decision threshold set from the business costs of each error; the random forest catches 95% of churners on unseen customers (ROC AUC 0.98).
- Ranking customers by expected loss protects 82% of the value at risk at 10% contact capacity, against 58% when ranking by churn probability; exported a ranked call list with a spending ceiling per customer.
- Built and deployed a Streamlit planner where a retention manager sets capacity, costs and save rate and gets the break-even call list, value protected and expected net value, using out-of-fold scores for all 2,850 customers.

A/B Test: New Landing Page — Python, statsmodels, SciPy

github.com/vusumuzinkosi/ab-test-landing-page

- Pre-registered a two-sided test (α = 0.05, +0.5-point minimum worthwhile lift) on 290,584 visitors, after cleaning mismatched rows and duplicate users.
- Recommended not rolling out: the new page converted 0.16 points lower (p = 0.19, 95% CI −0.39 to +0.08), with 98% power for the lift that matters.

Customer Segmentation: RFM and K-means Python, SQL (DuckDB), scikit-learn

github.com/vusumuzinkosi/customer-segmentation-rfm

- Segmented 5,852 customers from two years of retail transactions with hand-built RFM tiers, then K-means on log-scaled RFM with k chosen by silhouette score and business use.
- Found loyal big spenders are 19% of customers but 72% of revenue; segments built on older data predicted the next six months, with the top group spending 19× as much as lapsed buyers.
- Rebuilt the cleaning and RFM pipeline in SQL with DuckDB (CTEs, window functions, cohort retention), matching pandas exactly on 776,577 lines; found only about 1 in 5 new customers reorders within a month.

WORK EXPERIENCE

Senior Lecturer, Mathematics & IT Feb 2017 – Present

Majuba TVET College, Newcastle, KwaZulu-Natal

- Analyse learner performance data to identify at-risk students early and target evidence-based interventions, improving pass rates.
- Teach mathematics and IT to 100+ students a year, designing assessments and curriculum delivery aligned with national standards.
- Explain quantitative concepts to non-specialist audiences every day, the same skill used to present model results to business stakeholders.
- Developed standardised assessment rubrics adopted department-wide, and mentored IT competition teams to national rounds.

Special Educator, Mathematics Jan 2012 – Mar 2016

Mdumiseni High School, KwaZulu-Natal

- Taught mathematics to diverse learner groups, tracking individual progress and adapting materials to different learning needs.

EDUCATION

Postgraduate Diploma in Data Science (NQF 8) 2026 (in progress)

STADIO Higher Education, Remote

- 120-credit qualification (SAQA ID: 122721); modules: Introduction to Data Science, Working with Data, Statistical Modelling, Applied Machine Learning, Capstone Project.

Data Science Programme Mar 2025 – Apr 2026

ALX Africa, Remote

Postgraduate Diploma in TVET 2024 – 2025

University of the Western Cape

BSc Mathematics & Computer Science 2019 – 2024

University of South Africa

BEd Mathematics & Computer Science 2012 – 2017

University of South Africa

CERTIFICATIONS

- Data Science Certificate, ALX Africa (2025–2026)
- Data Analyst Certificate, ALX Africa (2025): SQL, Power BI, A/B Testing, Regression Analysis
- AI Engineering Bootcamp, Zero To Mastery Academy (2025)
- Prompt Engineering Bootcamp, Zero To Mastery Academy (2025)